SOP-108 — Control Valve Calibration and Stroke Testing

Plant Site A · Rev 2 · Issued 19/02/2025 · Approved by Plant Engineering

1. Purpose

This procedure defines the method for functional checking, stroke testing and calibration of the pneumatically actuated control valves at Plant Site A. Accurate valve response underpins stable process control, and a drifting positioner or sticking stem is a common hidden cause of process upsets that are wrongly blamed on upstream equipment.

2. Scope

Applies to control valves CV-301A and CV-302A in the Process Area and CV-303A in the Utilities Area, including their positioners, air sets and limit switches. Safety shutdown valves and relief devices are explicitly excluded and are covered by the safety instrumented system test procedures.

3. References

SOP-118 Equipment Isolation and Lock-Out Tag-Out.

Valve data sheet and positioner calibration manual for the tag under test.

Instrument loop drawing for the associated control loop.

4. Safety Precautions

Agree the test with the panel operator before moving any valve. A valve stroked without warning can trip the unit or divert flow unexpectedly.

Where the valve must be tested in line, confirm the bypass is available and the process consequence of full stroking has been assessed by the shift supervisor.

Keep hands clear of the yoke and stem at all times. Actuator spring force is sufficient to sever a finger, and the valve may move without warning while connected to its controller.

5. Tools and Materials

Have available: a calibrated signal source or agreed use of the panel output, a calibrated gauge set for supply air and positioner output, the valve data sheet and positioner manual for the tag, packing gland spanners, a stopwatch for fail action timing, and the calibration record sheets. All test equipment shall carry current calibration labels traceable to the site standard.

Verify the valve tag plate against the work order and the loop drawing before touching anything. Adjacent valves in the rack share actuator types, and stroking the wrong valve upsets a healthy loop.

Confirm the panel operator is in continuous radio contact for the duration of the test. If contact is lost mid stroke, return the valve to its last agreed position and suspend the test until contact is restored.

6. Procedure

6.1 Place the control loop in manual at the panel and confirm with the panel operator that the process is lined up to tolerate valve movement, or that the valve is isolated with the bypass carrying flow.

6.2 Record the as-found condition: supply air pressure at the air set, positioner input and output gauge readings, and the valve position indicated against the panel output.

6.3 Check the air supply quality at the air set drain. Water or oil discharged at the drain indicates instrument air contamination and shall be reported, because it will drift every positioner on the header, not only this one.

6.4 Apply control signals of zero, twenty five, fifty, seventy five and one hundred percent from the panel or a signal source, and record the actual stem travel at each point in both the rising and falling directions.

6.5 Compute the hysteresis as the largest difference between rising and falling travel at the same signal. Hysteresis above two percent of span indicates packing friction or a worn positioner linkage and shall be corrected, not calibrated around.

6.6 Inspect the stem and packing area for leakage, scoring and paint transfer that indicates rubbing. Adjust packing gland nuts only enough to stop leakage; over-tightening is the usual cause of stiction. If stiction cannot be corrected by packing adjustment, record the observed stick slip amplitude and raise the valve for workshop overhaul rather than compensating by widening the controller dead band.

6.7 Verify the positioner zero and span: the valve shall just reach its seat at zero percent signal and just reach full travel at one hundred percent. Adjust per the positioner manual and repeat the five-point check after any adjustment.

6.8 Confirm the failure action by isolating the air supply and observing the valve move to its specified fail position within the expected time. Restore the air supply afterwards.

6.9 Exercise the limit switches, where fitted, and confirm the open and closed indications display correctly at the panel. Where switch indication disagrees with the observed valve position, correct the switch setting rather than the panel graphic, and record the as-found discrepancy on the calibration sheet.

6.10 Return the loop to automatic in coordination with the panel operator and observe the loop for five minutes to confirm stable control at the normal setpoint.

7. Acceptance Criteria

The valve is accepted when travel error at every check point is within one percent of span, hysteresis is within two percent, the fail action operates correctly, and the loop holds a steady setpoint in automatic. Any valve failing these criteria shall be reported with the as-found data on the work order for trending.

8. Records

Record as-found and as-left five-point data, hysteresis, air set pressure and fail action result on the calibration sheet, and file against the valve tag. Trends of successive calibrations identify valves that are wearing and drive overhaul decisions. Records shall additionally be reviewed whenever the associated control loop shows sustained oscillation or steady state offset, before any controller retuning is attempted.